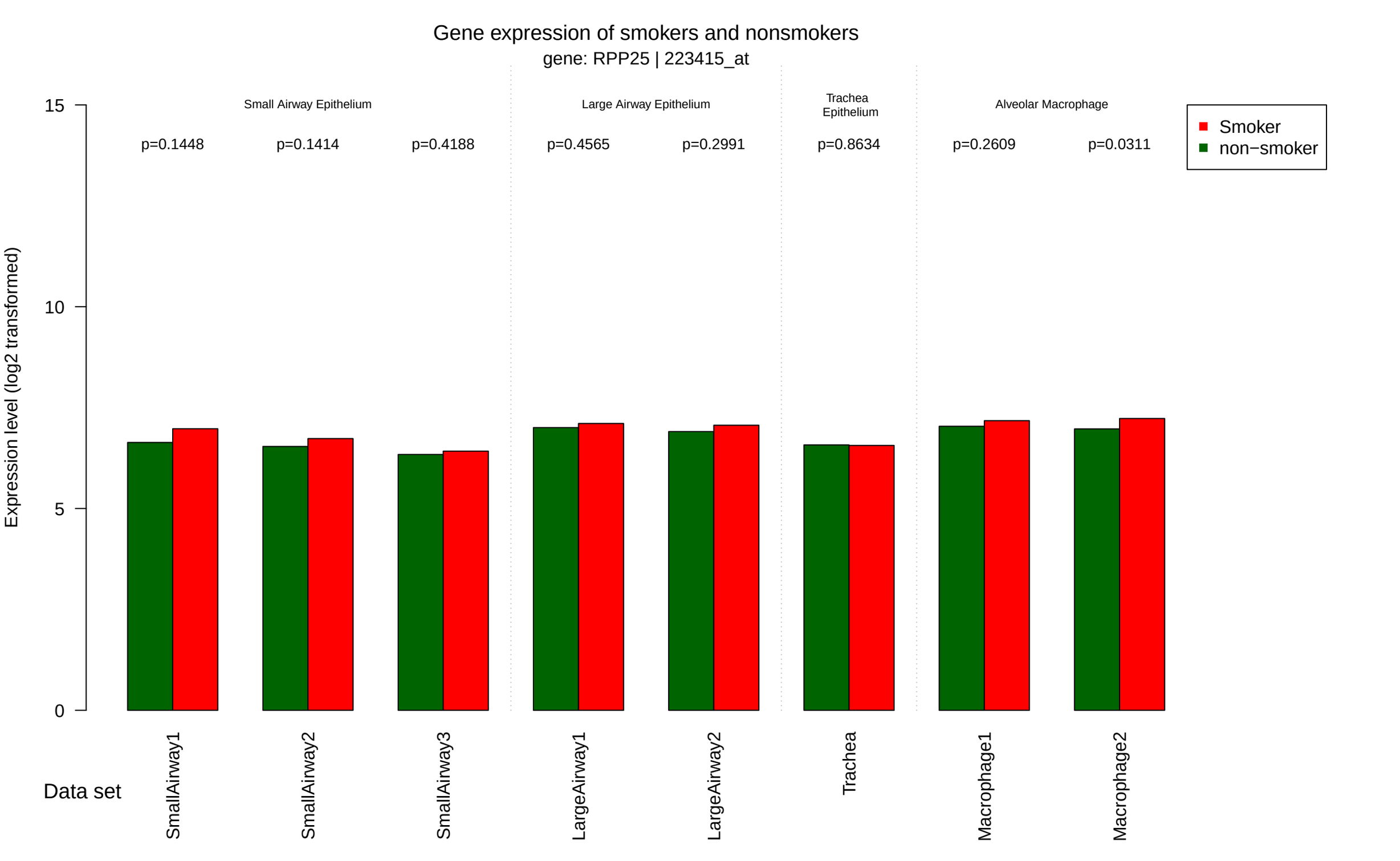
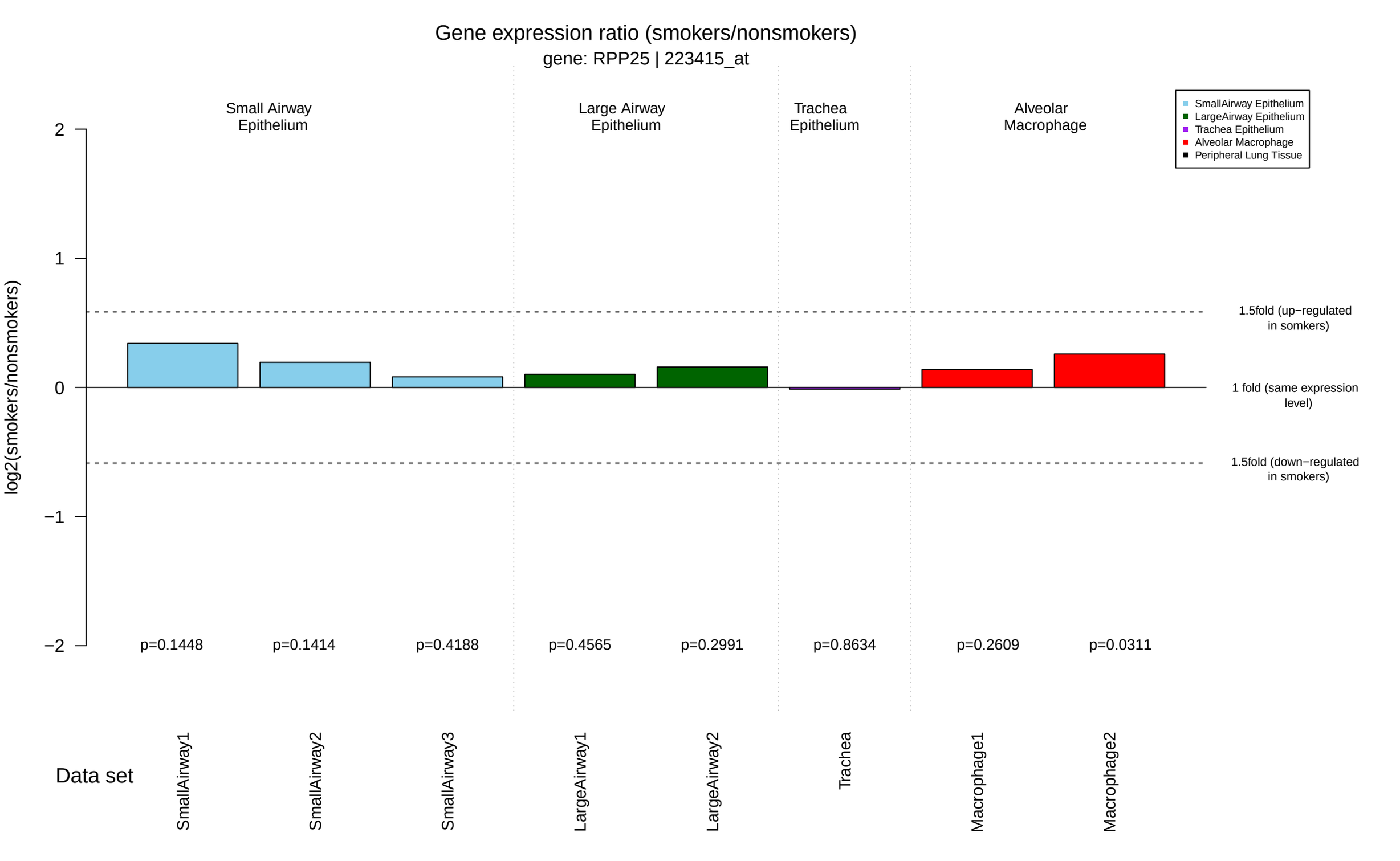
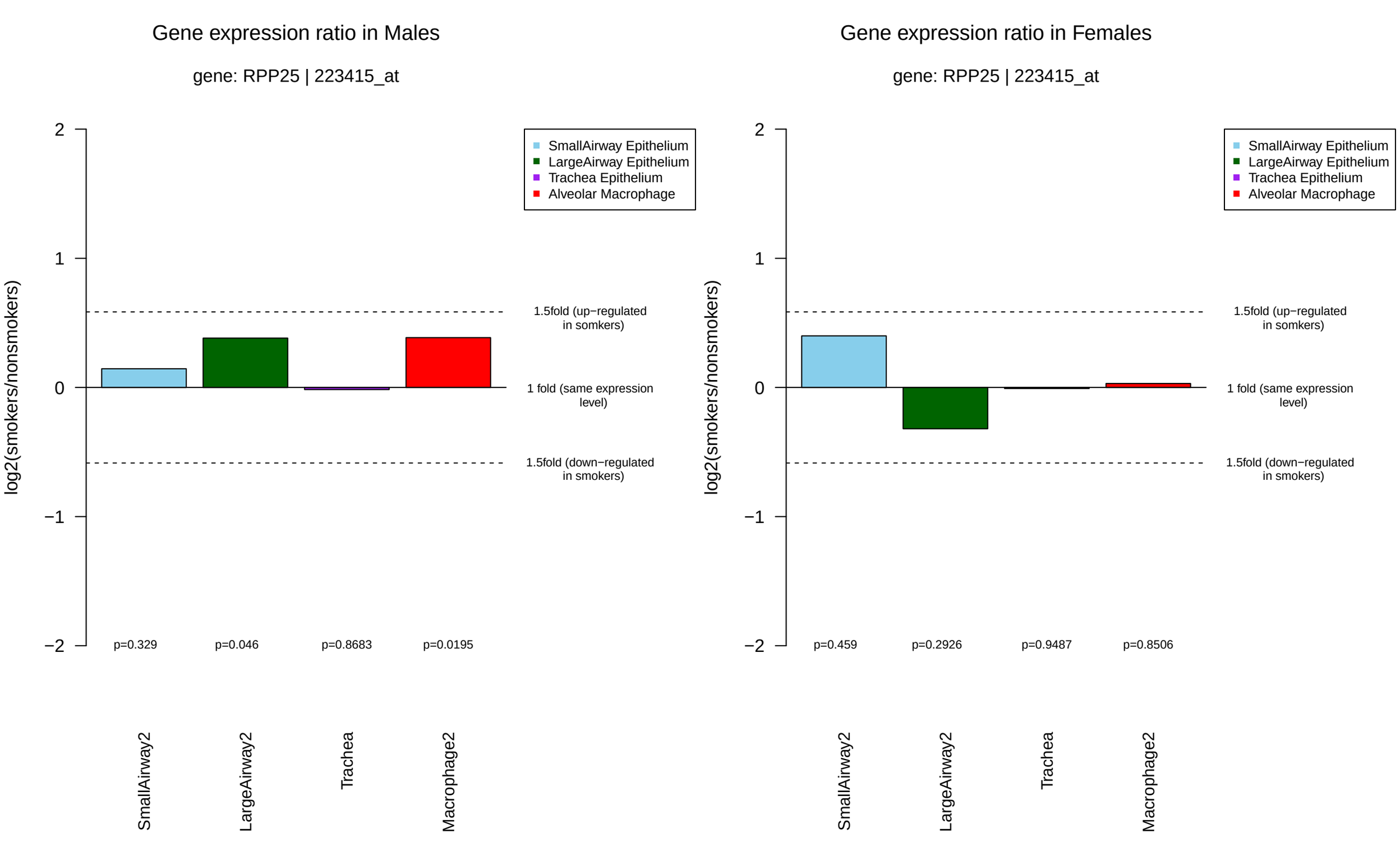
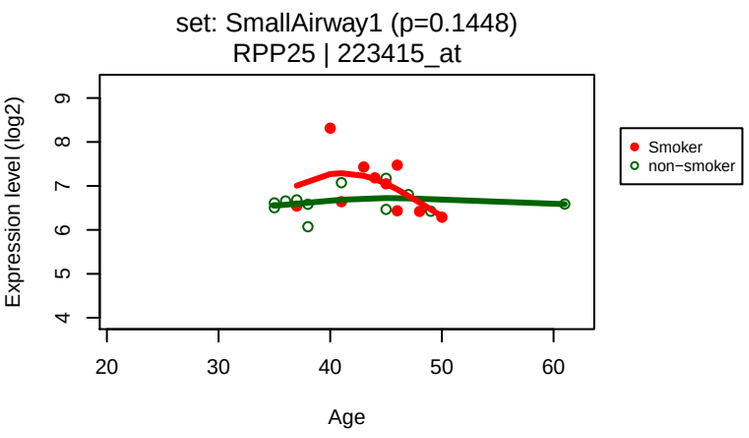
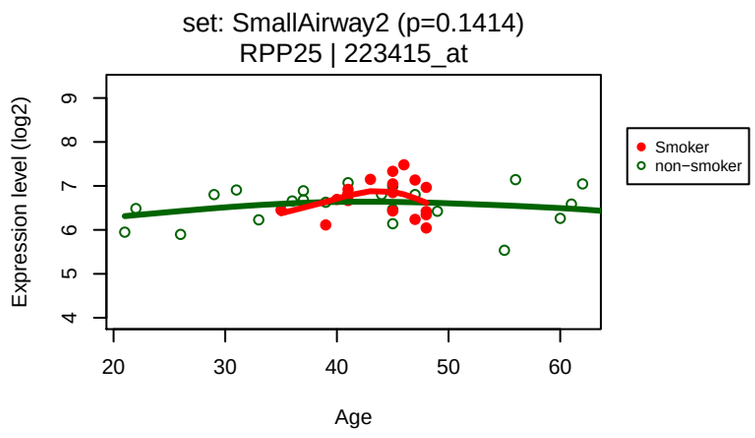
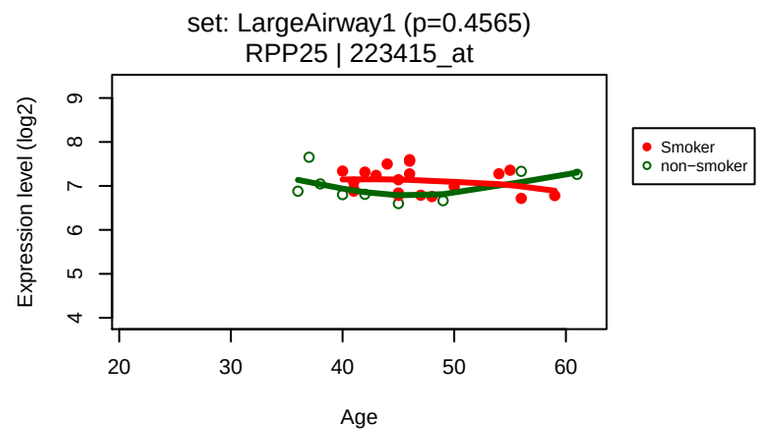

Gene expression of smokers and nonsmokers
gene: RPP25 | 223415_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1448
p=0.1414
p=0.4188
p=0.4565
p=0.2991
p=0.8634
p=0.2609
p=0.0311
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

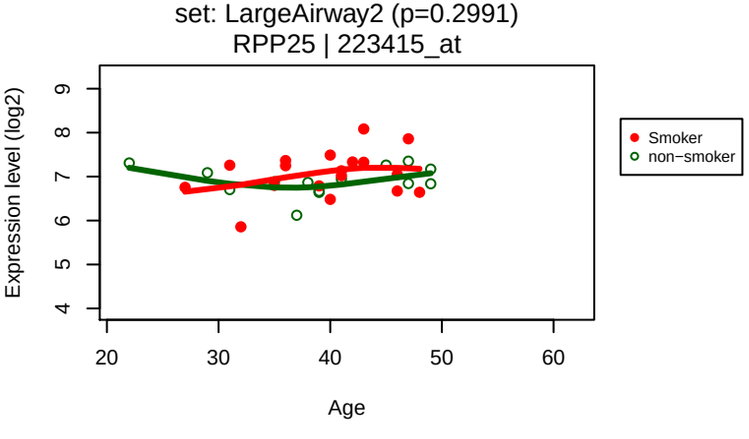

Gene expression ratio (smokers/nonsmokers)
gene: RPP25 | 223415_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1448
p=0.1414
p=0.4188
p=0.4565
p=0.2991
p=0.8634
p=0.2609
p=0.0311
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

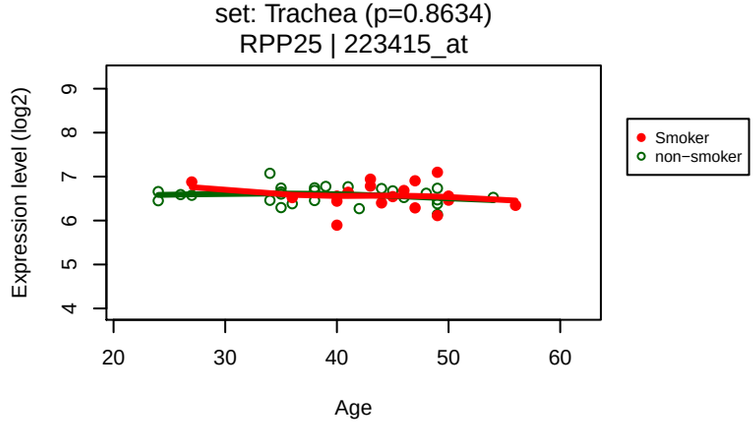

Gene expression ratio in Males
gene: RPP25 | 223415_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.329
p=0.046
p=0.8683
p=0.0195
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPP25 | 223415_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.459
p=0.2926
p=0.9487
p=0.8506
SmallAirway2
LargeAirway2
Trachea
Macrophage2

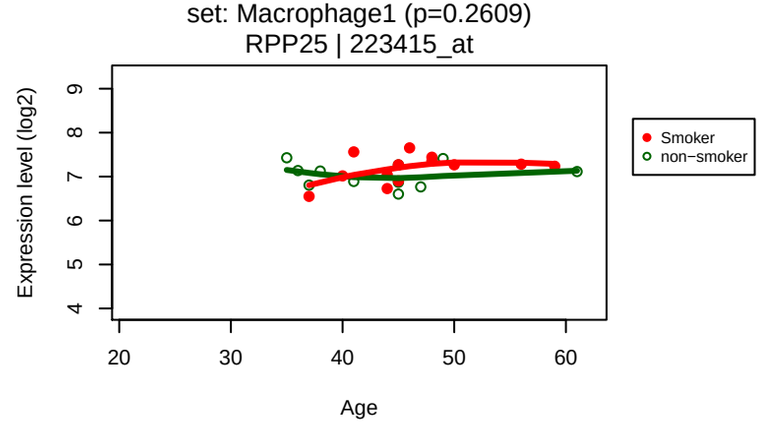

set: SmallAirway1 (p=0.1448)
RPP25 | 223415_at
Expression level (log2)
Age
Smoker
non−smoker

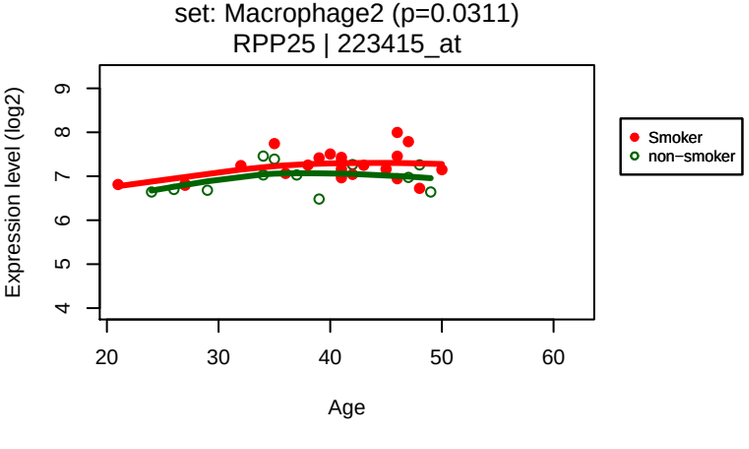

set: SmallAirway2 (p=0.1414)
RPP25 | 223415_at
Expression level (log2)
Age
Smoker
non−smoker

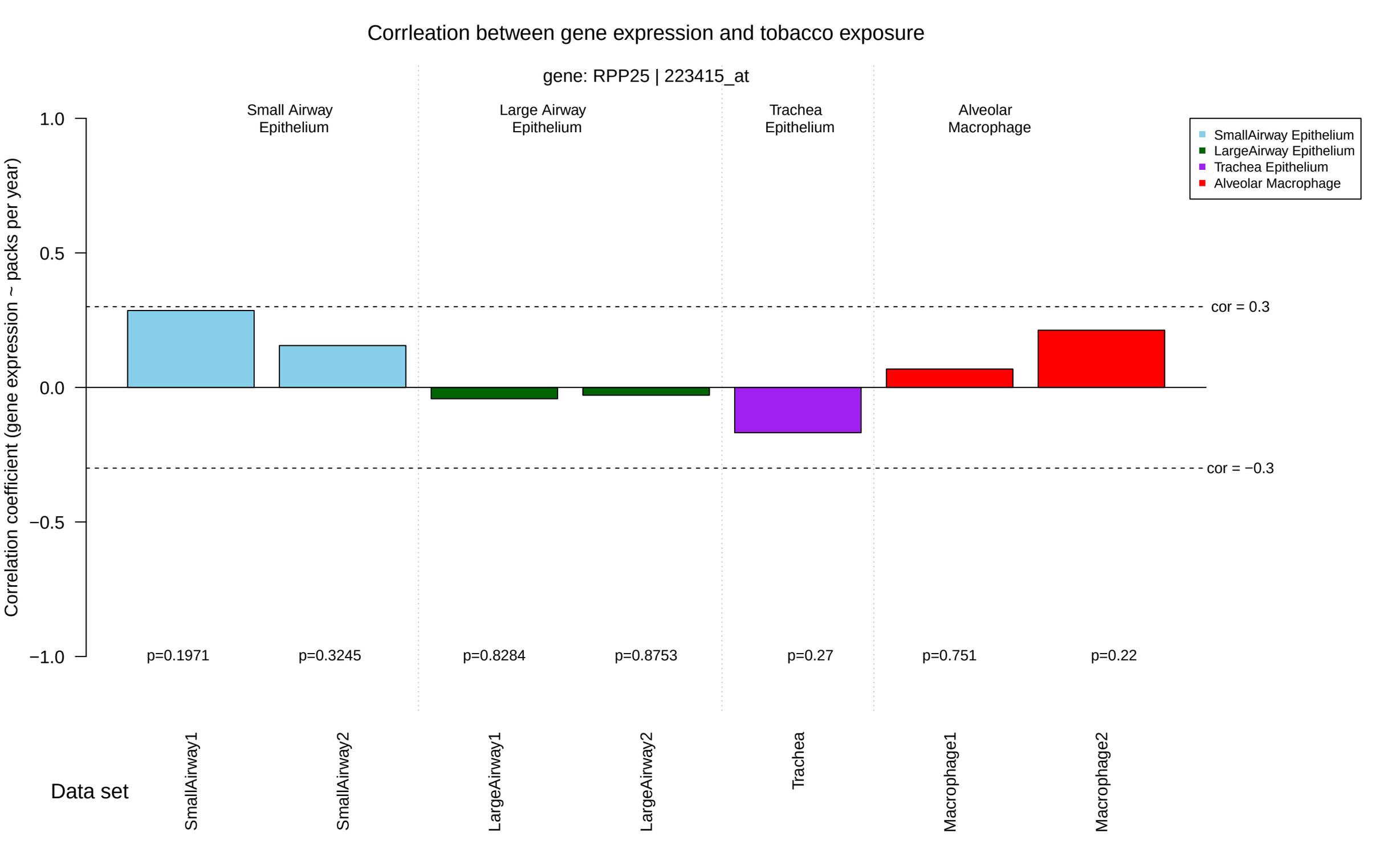

set: LargeAirway1 (p=0.4565)
RPP25 | 223415_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2991)
RPP25 | 223415_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8634)
RPP25 | 223415_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2609)
RPP25 | 223415_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0311)
RPP25 | 223415_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPP25 | 223415_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1971
p=0.3245
p=0.8284
p=0.8753
p=0.27
p=0.751
p=0.22
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2